# Chandan Kumar

Forward Deployed AI Engineer | Agentic AI · Full-Stack · AI Security

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## Summary

Engineer who takes a messy, real-world problem from first conversation to a product people use. **4+ years** building backend and distributed systems for banking and aviation; now on **Agentic AI and AI Security at Deloitte USI**. Built **AstroIntel**, a live multi-agent AI product, and **Bench Resource Optimizer** solo, measured on outcomes: success rate, cost per successful task and model evals. **262 pull requests merged** into open-source libraries such as pypdf and fontTools.

## Experience

**Deloitte USI** | *Software Engineer II, Agentic AI & AI Security* Sep 2026 – Present | *Bengaluru*

- Build agentic AI capabilities on an enterprise agent platform used for engineering and security workflows.
- Focus on fitting agents into existing workflows, making them reliable, and adopting AI securely.

**Infosys — Bank of America** | *Senior Software Engineer* Nov 2025 – Sep 2026 | *Pune*

- Integrated LLM APIs into Spring Boot services and built event-driven AI microservices on Kafka in a regulated banking environment, improving system efficiency by 40%.

**Nexsys — Accelya** | *Software Engineer* Dec 2023 – Oct 2025 | *Mumbai*

- Built and scaled Java microservices for airline systems handling 500K+ transactions a day.

**Texala** | *Software Engineer* May 2023 – Dec 2023 | *Pune*

- Built production microservices and enterprise REST APIs.

**Flyboard Ventures** | *Software Engineer* Aug 2022 – May 2023 | *Chandigarh*

- Shipped e-commerce, healthcare and e-learning products end to end, including APIs serving thousands of concurrent users.

## AI Projects — each scoped, built, shipped and run solo

**AstroIntel (Aura with Rav)** *Live · LangGraph, FastAPI, Angular, PostgreSQL, Docker*

- Multi-tenant AI product: LangGraph runs five domain agents in parallel and merges them into consensus insights; prompt-injection, hallucination and PII checks, and a human approves each insight before a report is generated.
- Every task stored with status, latency, tokens, retries and cost → success rate and **cost per successful task** dashboard; agent permission matrix (read / write / human approval, default deny) enforced in code.
- Reproducible model comparison on the production prompt: deepseek-chat matched deepseek-reasoner (6/6 cases, incl. fear-inducing and absolute claims) at about 14× lower cost.

**Bench Resource Optimizer** | *enterprise staffing* *Hybrid RAG, multi-agent, FastAPI, Angular*

- Matches bench engineers to project roles: a CV-parsing agent, hybrid RAG (FAISS + BM25 + RRF, HyDE, cross-encoder rerank), match % with skill gaps, then a day-by-day upskilling plan.
- Semantic cache, circuit breakers, PII filter and evals (incl. a prompt-injection CV); the outcome ledger exposed under-reported plan cost (tokens lost in worker threads), now fixed and tracked per task.

## Open Source

**262 merged pull requests across 24 libraries**, each a bug fix with a regression test, merged by maintainers. See all

- **pypdf** (82): hardened PDF parsing against malformed files: crash fixes, text extraction and image decoding.
- **Pillow** (13.9k⋆): invert, posterize and solarize for palette images in ImageOps, previously "coming soon".
- **txtai** (13k⋆): lists and paragraphs inside blockquotes kept intact in HTML-to-Markdown.
- **kornia** (11.4k⋆): unsharp_mask works on uint8 images · **tox** (4k⋆): env files accept `export KEY=value`, like dotenv.
- **fontTools** (7), **fpdf2** (89), **trafilatura**, **Authlib**, **Joblib**, **pydantic-extra-types**: clear errors for malformed fonts, browser-matching HTML/SVG rendering, input validation.

## Skills

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|---|---|
| **AI & Agents** | LLMs, LangGraph, LangChain, multi-agent systems, tool calling, RAG, LLM evals, model comparison |
| **Backend** | Python, Java, FastAPI, Spring Boot, Kafka, REST, SSE / WebSockets, PostgreSQL, Redis |
| **Frontend** | TypeScript, Angular, React |
| **Cloud & Ops** | Docker, Kubernetes, AWS, Vercel, Render, CI/CD, observability |
| **AI Security** | Prompt-injection defence, PII protection, output validation, guardrails, multi-tenant auth (JWT / RBAC) |

## Education

**Future Institute of Engineering and Management** | *B.Tech* 2014 – 2018 | *CGPA: 8.7*